1) a)

$$\sum_{r=1}^{30} 8 = 8 \times 30 = 240$$

1) b)

$$\sum_{r=1}^{50} r = \frac{1}{2} \times 50(50+1) = 1275$$

1) c)

$$\sum_{r=1}^{20} r^2 = \frac{1}{6} \times 20(20+1)(40+1) = 2870$$

1) d)

$$\sum_{r=1}^{10} r^3 = \frac{1}{4} \times 10^2(10+1)^2 = 3025$$

2) a)

$$\sum_{r=1}^{n} (2r+3) = 2\sum_{r=1}^{n} r + 3\sum_{r=1}^{n} 1 = 2 \times \frac{1}{2} n(n+1) + 3n = n^2 + 4n$$

2) b)

$$\sum_{r=1}^{n} (5r-12) = 5\sum_{r=1}^{n} r - 12\sum_{r=1}^{n} 1 = 5 \times \frac{1}{2} n(n+1) - 12n = \frac{5}{2} n^2 - \frac{19}{2} n$$

2) c)

$$\sum_{r=1}^{n} (4r+1) = 4\sum_{r=1}^{n} r + \sum_{r=1}^{n} 1 = 4 \times \frac{1}{2} n(n+1) + n = 2n^2 + 3n$$

2) d)

$$\sum_{r=1}^{n} (ar+b) = a\sum_{r=1}^{n} r + b\sum_{r=1}^{n} 1 = \frac{1}{2} an(n+1) + bn = \frac{1}{2} an^2 + \frac{1}{2} an + bn$$

3)

$$\sum_{r=1}^{k}(2r+k) = 2\sum_{r=1}^{k} r + k\sum_{r=1}^{k} 1 = 2\times\frac{1}{2}k(k+1) + k^2 = 2k^2 + k$$

$2k^2 + k = 300 \Rightarrow 2k^2 + k - 300 = 0$ which gives $k = -\frac{25}{2}$ and $k = 12$

$\therefore k = 12$

4)

$$\sum_{r=1}^{12} r(6r-1) = 6\sum_{r=1}^{12} r^2 - \sum_{r=1}^{12} r$$

$= 6\times\frac{1}{6}\times 12(12+1)(24+1) - \frac{1}{2}\times 12\times(12+1)$

$= 3822$ as req.

5)

$$\sum_{r=1}^{50}(3r+2)^2 = \sum_{r=1}^{50}(9r^2+12r+4) = 9\sum_{r=1}^{50} r^2 + 12\sum_{r=1}^{50} r + 4\sum_{r=1}^{50} 1$$

$$= 9\times\frac{1}{6}\times 50(50+1)(100+1) + 12\times\frac{1}{2}\times 50\times 51 + 4\times 50 = 401\,825$$

6) a)

$n(n+1) + 4n$

$= n[n+1+4]$

$= n(n+5)$

6) b)

$\frac{7}{2}n(n+1) - 5n$

$= \frac{1}{2}n[7(n+1) - 10]$

$= \frac{1}{2}n[7n+7-10]$

$= \frac{1}{2}n(7n-3)$

6) c)

$3n(n+1)(2n+1)+\frac{1}{2}n(n+1)$

$=\frac{1}{2}n(n+1)[6(2n+1)+1]$

$=\frac{1}{2}n(n+1)(12n+7)$

6) d)

$4n(n+1)(2n+1)-\frac{1}{4}n^2(n+1)^2$

$=\frac{1}{4}n(n+1)[16(2n+1)-n(n+1)]$

$=\frac{1}{4}n(n+1)[32n+16-n^2-n]$

$=\frac{1}{4}n(n+1)[-n^2+31n+16]$

6) e)

$\frac{1}{2}n^2(n+1)^2-3n(n+1)$

$=\frac{1}{2}n(n+1)[n(n+1)-6]$

$=\frac{1}{2}n(n+1)[n^2+n-6]$

$=\frac{1}{2}n(n+1)(n+3)(n-2)$

6) f)

$\frac{1}{4}n^2(n+1)^2-\frac{1}{6}n(n+1)(2n+1)$

$=\frac{1}{12}n(n+1)[3n(n+1)-2(2n+1)]$

$=\frac{1}{12}(n+1)[3n^2+3n-4n-2]$

$=\frac{1}{12}(n+1)[3n^2-n-2]$

$=\frac{1}{12}(n+1)(3n+2)(n-1)$

6) g)

$$\frac{3}{4}n^2(n+1)^2+\frac{5}{2}n(n+1)$$
$$=\frac{1}{4}n(n+1)[3n(n+1)+10]$$
$$=\frac{1}{4}n(n+1)(3n^2+3n+10)$$

7) a)

$$\sum_{r=1}^{n}(r^3+r^2)=\frac{1}{4}n^2(n+1)^2+\frac{1}{6}n(n+1)(2n+1)$$
$$=\frac{1}{12}n(n+1)[3n(n+1)+2(2n+1)]$$
$$=\frac{1}{12}n(n+1)[3n^2+3n+4n+2]$$
$$=\frac{1}{12}n(n+1)(3n^2+7n+2)$$
$$=\tfrac{1}{12}n(n+1)(3n+1)(n+2)$$

7) b)

$$\sum_{r=1}^{n}(2r^2+5r)=2\sum_{r=1}^{n}r^2+5\sum_{r=1}^{n}r$$
$$=2\times\frac{1}{6}n(n+1)(2n+1)+5\times\frac{1}{2}n(n+1)$$
$$=\frac{1}{6}n(n+1)[2(2n+1)+15]$$
$$=\frac{1}{6}n(n+1)(4n+17)$$

7) c)

$$\sum_{r=1}^{n}(7r^3-2r^2)=7\sum_{r=1}^{n}r^3-2\sum_{r=1}^{n}r^2$$
$$=7\times\frac{1}{4}n^2(n+1)-2\times\frac{1}{6}n(n+1)(2n+1)$$
$$=\frac{1}{12}n(n+1)[21n(n+1)-4(2n+1)]$$
$$=\frac{1}{12}n(n+1)[21n^2+21n-8n-4]$$
$$=\frac{1}{12}n(n+1)(21n^2+13n-4)$$

7) d)

$$\sum_{r=1}^{n}(1-2r+5r^3)=\sum_{r=1}^{n}1-2\sum_{r=1}^{n}r+5\sum_{r=1}^{n}r^3$$
$$=n-2\times\frac{1}{2}n(n+1)+5\times\frac{1}{4}n^2(n+1)^2$$
$$=\frac{1}{4}n[4-4(n+1)+5n(n+1)^2]$$
$$=\frac{1}{4}n[4-4n-4+5n(n^2+2n+1)]$$
$$=\frac{1}{4}n[-4n+5n^3+10n^2+5n]$$
$$=\frac{1}{4}n^2(5n^2+10n+1)$$

7) e)

$$\sum_{r=1}^{n}r(2r+1)=2\sum_{r=1}^{n}r^2+\sum_{r=1}^{n}r$$
$$=2\times\frac{1}{6}n(n+1)(2n+1)+\frac{1}{2}n(n+1)$$
$$=\frac{1}{6}n(n+1)[2(2n+1)+3]$$
$$=\frac{1}{6}n(n+1)(4n+5)$$

7) f)

$$\sum_{r=1}^{n} r^2(r-4) = \sum_{r=1}^{n} r^3 - 4\sum_{r=1}^{n} r^2$$

$$= \frac{1}{4}n^2(n+1)^2 - 4 \times \frac{1}{6}n(n+1)(2n+1)$$

$$= \frac{1}{12}n(n+1)[3n(n+1) - 8(2n+1)]$$

$$= \frac{1}{12}n(n+1)[3n^2 + 3n - 16n - 8]$$

$$= \frac{1}{12}n(n+1)(3n^2 - 13n - 8)$$

8)

$$\sum_{r=1}^{n} (4r-2)^2 = \sum_{r=1}^{n} (16r^2 - 16r + 4) = 16\sum_{r=1}^{n} r^2 - 16\sum_{r=1}^{n} r + 4\sum_{r=1}^{n} 1$$

$$= 4\left[4\sum_{r=1}^{n} r^2 - 4\sum_{r=1}^{n} r + \sum_{r=1}^{n} 1\right]$$

$$= 4\left[4 \times \frac{1}{6}n(n+1)(2n+1) - 4 \times \frac{1}{2}n(n+1) + n\right]$$

$$= \frac{4}{3}n[2(n+1)(2n+1) - 6(n+1) + 3]$$

$$= \frac{4}{3}n[2(2n^2 + 3n + 1) - 6(n+1) + 3]$$

$$= \frac{4}{3}n[4n^2 + 6n + 2 - 6n - 6 + 3]$$

$$= \frac{4}{3}n[4n^2 - 1] = \frac{4}{3}n(2n-1)(2n+1)$$

9) a)

$$(1\times 2)+(2\times 3)+(3\times 4)+\cdots=\sum_{r=1}^{n} r(r+1)$$

$$=\sum_{r=1}^{n} r^2+\sum_{r=1}^{n} r$$

$$=\frac{1}{6}n(n+1)(2n+1)+\frac{1}{2}n(n+1)$$

$$=\frac{1}{6}n(n+1)[2n+1+3]$$

$$=\frac{1}{3}n(n+1)(n+2)$$

9) b)

$$(3\times 5)+(4\times 6)+(5\times 7)+\cdots=\sum_{r=1}^{n}(r+2)(r+4)$$

$$=\sum_{r=1}^{n}(r^2+6r+8)$$

$$=\sum_{r=1}^{n} r^2+6\sum_{r=1}^{n} r+8\sum_{r=1}^{n} 1$$

$$=\frac{1}{6}n(n+1)(2n+1)+6\times\frac{1}{2}n(n+1)+8n$$

$$=\frac{1}{6}n[(n+1)(2n+1)+18(n+1)+48]$$

$$=\frac{1}{6}n[2n^2+3n+1+18n+18+48]$$

$$=\frac{1}{6}n(2n^2+21n+67)$$

9) c)

$$(2\times 4)+(3\times 5)+(4\times 6)+\cdots=\sum_{r=1}^{n}(r+1)(r+3)$$

$$=\sum_{r=1}^{n}(r^2+4r+3)$$

$$=\sum_{r=1}^{n}r^2+4\sum_{r=1}^{n}r+3\sum_{r=1}^{n}1$$

$$=\frac{1}{6}n(n+1)(2n+1)+4\times\frac{1}{2}n(n+1)+3n$$

$$=\frac{1}{6}n[(n+1)(2n+1)+12(n+1)+18]$$

$$=\frac{1}{6}n[2n^2+3n+1+12n+12+18]$$

$$=\frac{1}{6}n(2n^2+15n+31)$$

9) d)

$$2\times 1^2+3\times 2^2+4\times 3^2+\cdots=\sum_{r=1}^{n}(r+1)r^2$$

$$=\sum_{r=1}^{n}(r^3+r^2)$$

$$=\sum_{r=1}^{n}r^3+\sum_{r=1}^{n}r^2$$

$$=\frac{1}{4}n^2(n+1)^2+\frac{1}{6}n(n+1)(2n+1)$$

$$=\frac{1}{12}n(n+1)[3n(n+1)+2(2n+1)]$$

$$=\frac{1}{12}n(n+1)[3n^2+3n+4n+2]$$

$$=\frac{1}{12}n(n+1)[3n^2+7n+2]$$

$$=\frac{1}{12}n(n+1)(3n+1)(n+2)$$

9) e)

$$(1\times3\times4)+(2\times4\times5)+(3\times5\times6)+\cdots=\sum_{r=1}^{n}r\,(r+2)(r+3)$$

$$=\sum_{r=1}^{n}(r^3+5r^2+6r)$$

$$=\sum_{r=1}^{n}r^3+5\sum_{r=1}^{n}r^2+6\sum_{r=1}^{n}r$$

$$=\frac{1}{4}n^2(n+1)^2+5\times\frac{1}{6}n(n+1)(2n+1)+6\times\frac{1}{2}n(n+1)$$

$$=\frac{1}{12}n(n+1)[3n(n+1)+10(2n+1)+36]$$

$$=\frac{1}{12}n(n+1)[3n^2+3n+20n+10+36]$$

$$=\frac{1}{12}n(n+1)(3n^2+23n+46)$$

10) a)

$$\sum_{r=1}^{n}(r+1)(3r-1)=\sum_{r=1}^{n}(3r^2+2r-1)$$

$$=3\sum_{r=1}^{n}r^2+2\sum_{r=1}^{n}r-\sum_{r=1}^{n}1$$

$$=3\times\frac{1}{6}n(n+1)(2n+1)+2\times\frac{1}{2}n(n+1)-n$$

$$=\frac{1}{2}n[(n+1)(2n+1)+2(n+1)-2]$$

$$=\frac{1}{2}n[2n^2+3n+1+2n+2-2]$$

$$=\frac{1}{2}n(2n^2+5n+1)$$

10) b)

$2\times2+3\times5+4\times8+\cdots+51\times149$ is

$\sum_{r=1}^{n}(r+1)(3r-1)$ with $r=1$ and $n=50$

$\frac{1}{2}n(2n^2+5n+1)=\frac{1}{2}\times50(2\times50^2+5\times50+1)=131\,275$

11) a)

$$\sum_{r=1}^{2n} r^2 = \frac{1}{6}(2n)(2n+1)(2(2n)+1) = \frac{1}{3}n(2n+1)(4n+1)$$

11) b)

$$\sum_{r=1}^{2n} (2r)^2 = 4\sum_{r=1}^{2n} r^2 = \frac{4}{6}(2n)(2n+1)(2(2n)+1) = \frac{4}{3}n(2n+1)(4n+1)$$

11) c)

$$\sum_{r=1}^{2n} (2r-1)^2 = 4\sum_{r=1}^{2n} r^2 - 4\sum_{r=1}^{2n} r + \sum_{r=1}^{2n} 1$$

$$= \frac{4}{6}(2n)(2n+1)(2(2n)+1) - 4 \times \frac{1}{2}(2n)(2n+1) + 2n$$

$$= \frac{4}{3}n(2n+1)(4n+1) - 4n(2n+1) + 2n$$

$$= \frac{4}{3}n(2n+1)(4n+1) - 8n^2 - 2n$$

$$= \frac{4}{3}n(2n+1)(4n+1) - 2n(4n+1)$$

$$= \frac{2}{3}n(4n+1)[2(2n+1) - 3]$$

$$= \frac{2}{3}(4n+1)(4n-1)$$

12) a)

$$2^2 + 4^2 + 6^2 + \cdots + 20^2 = \sum_{r=1}^{10} (2r)^2 = 4\sum_{r=1}^{10} r^2$$

$$= 4 \times \frac{1}{6} \times 10 \times 11 \times 21$$

$= 1540$ as req.

12) b)

$$1^2 + 3^2 + 5^2 + \cdots + 19^2 = \sum_{r=1}^{10} (2r-1)^2 = \sum_{r=1}^{10} (4r^2 - 4r + 1)$$

$$= 4\sum_{r=1}^{10} r^2 - 4\sum_{r=1}^{10} r + \sum_{r=1}^{10} 1$$

$$= 4 \times \frac{1}{6} \times 10 \times 11 \times 21 - 4 \times \frac{1}{2} \times 10 \times 11 + 10$$

$= 1330$ as req.

12) c)

$$2^3 + 4^3 + 6^3 + \cdots + 20^3 = \sum_{r=1}^{10} (2r)^3 = 8\sum_{r=1}^{10} r^3$$

$$= 8 \times \frac{1}{4} \times 10^2 \times 11^2$$

$= 24\,200$ as req.

13)

$$1 + 2 + 3 + 4 + 5 + \cdots = \sum_{r=1}^{999} r = \frac{1}{2} \times 999(999 + 1) = 499\,500$$

The square numbers up to 999 are $1^2 + 2^2 + 3^2 + \cdots + 31^2$

$$1^2 + 2^2 + 3^2 + \cdots + 31^2 = \sum_{r=1}^{31} r^2 = \frac{1}{6} \times 31(31 + 1)(62 + 1) = 10\,416$$

$499\,500 - 10\,416 = 489\,084$

14)

$2^2 + 4^2 + 6^2 + \cdots + (2r)^2 = 82160$ where the series has n terms.

$$\sum_{r=1}^{n} (2r)^2 = 4\sum_{r=1}^{n} r^2$$

$$= 4 \times \frac{1}{6} n(n+1)(2n+1)$$

$$\frac{2}{3} n(n+1)(2n+1) = 82160$$

$$n(n+1)(2n+1) = 123240$$

$$2n^3 + 3n^2 + n = 123240$$

$$2n^3 + 3n^2 + n - 123240 = 0$$

which has the single real solution of $n = 39$

15)

$1^2 + 3^2 + 5^2 + \cdots + (2r-1)^2 = 20\,825$ where the series has n terms.

$$\sum_{r=1}^{n} (2r-1)^2 = \sum_{r=1}^{n} (4r^2 - 4r + 1) = 4\sum_{r=1}^{n} r^2 - 4\sum_{r=1}^{n} r + \sum_{r=1}^{n} 1$$

$$= 4 \times \frac{1}{6} n(n+1)(2n+1) - 4 \times \frac{1}{2} n(n+1) + n$$

$$\frac{2}{3} n(n+1)(2n+1) - 2n(n+1) + n = 20\,825$$

$$\frac{2}{3} n(2n^2 + 3n + 1) - 2n^2 - 2n + n = 20\,825$$

$$2n(2n^2 + 3n + 1) - 6n^2 - 6n + 3n = 62\,475$$

$$4n^3 + 6n^2 + 2n - 6n^2 - 6n + 3n = 62\,475$$

$$4n^3 - n = 62\,475$$

$$4n^3 - n - 62\,475 = 0$$

which has the single real solution of $n = 25$

16) a)

$$\sum_{r=1}^{n}(r+2)(r+3)=\sum_{r=1}^{n}(r^2+5r+6)=\sum_{r=1}^{n}r^2+5\sum_{r=1}^{n}r+6\sum_{r=1}^{n}1$$

$$=\frac{1}{6}n(n+1)(2n+1)+5\times\frac{1}{2}n(n+1)+6n$$

$$24+3\sum_{r=1}^{n}(r+2)(r+3)=24+\frac{1}{2}n(n+1)(2n+1)+\frac{15}{2}n(n+1)+18n$$

$$=24+\tfrac{1}{2}n(n+1)(2n+1)+\tfrac{15}{2}n(n+1)+18n$$

$$=24+n^3+\tfrac{3}{2}n^2+\tfrac{1}{2}n+\tfrac{15}{2}n^2+\tfrac{15}{2}n+18n$$

$$=n^3+9n^2+26n+24$$

$$=(n+2)(n^2+7n+12)$$

$$=(n+2)(n+3)(n+4)$$

16) b)

$$24+3\sum_{r=1}^{n}(r+2)(r+3)=(n+2)(n+3)(n+4)$$

which are consecutive integers, one of which must be divisible by 2 and another by 3 therefore $24+3\sum_{r=1}^{n}(r+2)(r+3)$ is divisible by 6

17) a)

$$\sum_{r=1}^{n}(7r-2)^2 = \sum_{r=1}^{n}(49r^2-28r+4)$$

$$= 49\sum_{r=1}^{n}r^2 - 28\sum_{r=1}^{n}r + 4\sum_{r=1}^{n}1$$

$$= 49\times\frac{1}{6}n(n+1)(2n+1) - 28\times\frac{1}{2}n(n+1) + 4n$$

$$= \frac{1}{6}n[49(n+1)(2n+1) - 84(n+1) + 24]$$

$$= \frac{1}{6}n[49(2n^2+3n+1) - 84n - 84 + 24]$$

$$= \frac{1}{6}n[98n^2+147n+49-84n-84+24]$$

$$= \frac{1}{6}n(98n^2+63n-11)$$

$a = 98$, $b = 63$ and $c = -11$

17) b)

$$\frac{1}{6}k(98k^2+63k-11) = 190k^2$$

$$98k^3+63k^2-11k = 1140k^2$$

$$98k^3-1077k^2-11k = 0$$

which has solutions $k = 0$, $k = -\frac{1}{98}$ and $k = 11$

$\therefore k = 11$

18) a)

$$\sum_{r=1}^{n}ar+b = a\sum_{r=1}^{n}r + \sum_{r=1}^{n}b = a\times\frac{1}{2}n(n+1) + bn$$

$a\times\frac{1}{2}\times 3\times 4 + 3b = 0 \Rightarrow 6a+3b = 0$

$a\times\frac{1}{2}\times 7\times 8 + 7b = 35 \Rightarrow 28a+7b = 35$

Gives $a = \frac{5}{2}$ and $b = -5$

So that $f(r) = \frac{5}{2}r - 5$

18) b)

$$\sum_{r=1}^{20} \frac{5}{2}r - 5 = \frac{5}{2} \times \frac{1}{2} \times 20 \times 21 - 5 \times 20 = 425$$

19) a)

$$\sum_{r=1}^{n} (5r - 7) = 5\sum_{r=1}^{n} r - \sum_{r=1}^{n} 7 = 5 \times \frac{1}{2}n(n+1) - 7n$$

$$= \frac{1}{2}n(5(n+1) - 14)$$

$$= \frac{1}{2}n(5n + 5 - 14)$$

$= \frac{1}{2}n(5n - 9)$ as req.

19) b)

$\frac{1}{2}n(5n - 9) > 6000$

$5n^2 - 9n - 12000 > 0$

$5n^2 - 9n - 12000 = 0$ has solutions $n = \frac{9}{10} \pm \frac{1}{10}\sqrt{240081}$

So $n = 50$

20)

$$\sum_{r=1}^{n} (ar + b) = a\sum_{r=1}^{n} r + bn$$

$$= a \times \frac{1}{2}n(n+1) + bn$$

$$= \frac{1}{2}an^2 + \frac{1}{2}an + bn$$

$$= \frac{1}{2}an^2 + \left(\frac{1}{2}a + b\right)n$$

$\frac{1}{2}a = 1$ so $a = 2$

$\frac{1}{2} \times 2 + b = -10$

$\therefore a = 2$ and $b = -11$

21) a)

$$\sum_{r=1}^{n}(r+10)(r+12)=\sum_{r=1}^{n}(r^2+22r+120)$$

$$=\sum_{r=1}^{n}r^2+22\sum_{r=1}^{n}r+120\sum_{r=1}^{n}1$$

$$=\frac{1}{6}n(n+1)(2n+1)+22\times\frac{1}{2}n(n+1)+120n$$

$$=\frac{1}{6}n[(n+1)(2n+1)+66(n+1)+720]$$

$$=\frac{1}{6}n[2n^2+3n+1+66n+66+720]$$

$$=\frac{1}{6}n(2n^2+69n+787)$$ as req.

21) b)

$$\frac{1}{6}n(2n^2+69n+787)=3806$$

$$2n^3+69n^2+787n=22836$$

$$2n^3+69n^2+787n-22836=0$$

The only real solution is $n=12$

22) a)

$$210+40r$$

22) b)

$$\left(50+\frac{r^2}{30}\right)(210+40r)-(2000+7r^2)$$

$$=10500+2000r+7r^2+\frac{4}{3}r^3-2000-7r^2$$

$$=8500+2000r+\frac{4}{3}r^3$$

$$=\frac{4}{3}(r^3+1500r+6375)$$ as req.

22) c)

$$\sum_{r=1}^{n}\frac{4}{3}(r^3+1500r+6375)=\frac{4}{3}\sum_{r=1}^{n}r^3+2000\sum_{r=1}^{n}r+8500\sum_{r=1}^{n}1$$

$$=\frac{4}{3}\times\frac{1}{4}n^2(n+1)^2+2000\times\frac{1}{2}n(n+1)+8500n$$

$$=\frac{1}{3}n[n(n+1)^2+3000(n+1)+25500]$$

$$=\frac{1}{3}n[n(n^2+2n+1)+3000n+3000+25500]$$

$$=\frac{1}{3}n(n^3+2n^2+3001n+28500)$$

22) d)

$\frac{1}{3}\times 15(15^3+2\times 15^2+3001\times 15+28500)=£386\ 700$

This result is much higher than the actual profit, so the model is over-estimating the profit and so it is not a good model.

23) a)

$397+3r$

23) b)

$$\sum_{r=1}^{n}397+3r=397\sum_{r=1}^{n}1+3\sum_{r=1}^{n}r$$

$$=397n+3\times\frac{1}{2}n(n+1)$$

$$=\frac{1}{2}n[794+3(n+1)]$$

$$=\frac{1}{2}n(3n+797)$$

There are n loaves so that

$\frac{1}{2}(3n+797)$ as req.

23) c)

Each loaf has mass $(397 + 3r)$

$$\sum_{r=1}^{20}(397 + 3r) = \frac{1}{2} \times 20(3 \times 20 + 797) = 8570$$

$$\sum_{r=1}^{20}(397 + 3r)^2 = \sum_{r=1}^{20}(9r^2 + 2382r + 157609)$$

$$= 9\sum_{r=1}^{20} r^2 + 2832\sum_{r=1}^{20} r + 157609\sum_{r=1}^{20} 1$$

$$= 9 \times \frac{1}{6} \times 20(20 + 1)(40 + 1) + 2382 \times \frac{1}{2} \times 20 \times 21 + 20 \times 157609$$

$$= 3678230$$

$$\sqrt{\frac{\sum r^2}{n} - \left(\frac{\sum r}{n}\right)^2} = \sqrt{\frac{3678230}{20} - \left(\frac{8570}{20}\right)^2} = \sqrt{183911.5 - 428.5^2} = 17.3 \text{ grams (3 SF)}$$

24) a)

$$1^3 + 2^3 + 3^3 + \cdots + n^3 = \sum_{r=1}^{n} r^3 = \frac{1}{4}n^2(n + 1)^2$$

$$1 + 2 + 3 + \cdots + n = \sum_{r=1}^{n} r = \frac{1}{2}n(n + 1)$$

$$\frac{1^3+2^3+3^3+\cdots+n^3}{1+2+3+\cdots+n} = \frac{\frac{1}{4}n^2(n+1)^2}{\frac{1}{2}n(n+1)} = \frac{1}{2}n(n + 1) \text{ as req.}$$

24) b)

$$\frac{1}{2}x(x + 1) > 200$$

$$\frac{1}{2}x^2 + \frac{1}{2}x - 200 > 0$$

$\frac{1}{2}x^2 + \frac{1}{2}x - 200 = 0$ has roots $x = -\frac{1}{2} \pm \frac{1}{2}\sqrt{1601}$

$x = 19.506 \ldots$

$\therefore n = 20$

25) a)

$$\sum_{r=1}^{n}(r^3-2r)=\sum_{r=1}^{n}r^3-2\sum_{r=1}^{n}r$$
$$=\frac{1}{4}n^2(n+1)^2-2\times\frac{1}{2}n(n+1)$$
$$=\frac{1}{4}n(n+1)[n(n+1)-4]$$
$$=\frac{1}{4}n(n+1)(n^2+n-4)$$

25) b)

$$\sum_{r=1}^{k}(2r^3+3r)-\sum_{r=1}^{k}(r^3+5r)=\sum_{r=1}^{k}(r^3-2r)=\frac{1}{4}n(n+1)(n^2+n-4)$$
$$\frac{1}{4}n(n+1)(n^2+n-4)=1224$$
$$n(n^3+2n^2-3n-4)=4896$$
$$n^4+2n^3-3n^2-4n-4896=0$$

which has real solutions $n=-9$ and $n=8\therefore k=8$

26) a)

$$\sum_{r=10-9}^{30-9}\big((r+9)-15\big)\big((r+9)-16\big)=\sum_{r=1}^{21}(r-6)(r-7)$$

26) b)

$$\sum_{r=4-3}^{16-3}\big((r+3)^2+3(r+3)\big)=\sum_{r=1}^{13}(r^2+9r+18)$$

26) c)

$$\sum_{r=12-11}^{20-11}(3(r+11)^3-1)=\sum_{r=1}^{9}(3(r+11)^3-1)$$

$$=\sum_{r=1}^{9}(3(r^3+33r^2+363r+1331)-1)$$

$$=\sum_{r=1}^{9}(3r^3+99r^2+1089r+3992)$$

26) d)

$$\sum_{r=1}^{8}r^2-\sum_{r=1}^{4}r^2=\sum_{r=1}^{4}r^2+\sum_{r=5}^{8}r^2-\sum_{r=1}^{4}r^2=\sum_{r=5}^{8}r^2=\sum_{r=5-4}^{8-4}(r+4)^2=\sum_{r=1}^{4}(r+4)^2$$

$$=\sum_{r=1}^{4}(r^2+8r+16)$$

27) a)

$$\sum_{r=10}^{20}(r^2-3)=\sum_{r=1}^{11}((r+9)^2-3)=\sum_{r=1}^{11}(r^2+18r+78)$$

$$=\sum_{r=1}^{11}r^2+18\sum_{r=1}^{11}r+78\sum_{r=1}^{11}1$$

$$=\frac{1}{6}\times 11(23)(12)+18\times\frac{11}{2}(12)+78\times 11$$

$$=2552$$

27) b)

$$\sum_{r=5}^{10}r(r-1)=\sum_{r=1}^{6}(r+4)(r+3)=\sum_{r=1}^{6}(r^2+7r+12)$$

$$=\sum_{r=1}^{6}r^2+7\sum_{r=1}^{6}r+12\sum_{r=1}^{6}1$$

$$=\frac{1}{6}\times 6(13)(7)+7\times\frac{6}{2}(7)+12\times 6$$

$$=310$$

27) c)

$$\sum_{r=50}^{100}(r^3-1)=\sum_{r=1}^{51}((r+49)^3-1)=\sum_{r=1}^{51}(r^3+147r^2+7203r+117648)$$
$$=\sum_{r=1}^{51}r^3+147\sum_{r=1}^{51}r^2+7203\sum_{r=1}^{51}r+117648\sum_{r=1}^{51}1$$
$$=\frac{1}{4}\times 51^2\times 52^2+147\times\frac{51}{6}\times 52\times 103+7203\times\frac{51}{2}\times 52+117648\times 51$$
$$=24\,001\,824$$

28)

$$51^2+52^2+53^2+\cdots+101^2=\sum_{r=51}^{101}r^2$$
$$=\sum_{r=1}^{51}(r+50)^2=\sum_{r=1}^{51}(r^2+100r+2500)=\sum_{r=1}^{51}r^2+100\sum_{r=1}^{51}r+2500\sum_{r=1}^{51}1$$
$$=\frac{1}{6}\times 51(51+1)(102+1)+100\times\frac{1}{2}\times 51(52)+2500\times 51$$
$$=305\,626$$

29) a)

$$\sum_{r=1}^{n}r(r-1)(r+1)=\sum_{r=1}^{n}r(r^2-1)=\sum_{r=1}^{n}(r^3-r)$$
$$=\sum_{r=1}^{n}r^3-\sum_{r=1}^{n}r$$
$$=\frac{1}{4}n^2(n+1)-\frac{1}{2}n(n+1)$$
$$=\frac{1}{4}n(n+1)[n(n+1)-2]$$
$$=\frac{1}{4}n(n+1)[n^2+n-2]$$
$$=\frac{1}{4}n(n+1)(n-1)(n+2)\text{ as req.}$$

29) b)

$$\sum_{r=1}^{2n} r(r-1)(r+1) = \frac{1}{4} \times 2n(2n+1)(2n-1)(2n+2)$$

$$= \frac{1}{2} \times 2n(2n+1)(2n-1)(n+1)$$

$$= n(2n+1)(2n-1)(n+1) \text{ as req.}$$

30) a)

$$\sum_{r=1}^{2n} r^3 = \frac{1}{4}(2n)^2(2n+1)^2 = n^2(2n+1)^2$$

30) b)

$$\sum_{r=n+1}^{2n} r^2 = \sum_{r=1}^{2n} r^2 - \sum_{r=1}^{n} r^2$$

$$= \frac{1}{6}(2n)(2n+1)(2(2n)+1) - \frac{1}{6}n(n+1)(2n+1)$$

$$= \frac{1}{3}n(2n+1)(4n+1) - \frac{1}{6}n(n+1)(2n+1)$$

$$= \frac{1}{6}n[2(2n+1)(4n+1) - (n+1)(2n+1)]$$

$$= \frac{1}{6}n[16n^2 + 12n + 2 - 2n^2 - 3n - 1]$$

$$= \frac{1}{6}n(14n^2 + 9n + 1)$$

$$= \frac{1}{6}n(7n+1)(2n+1)$$

Alternative Method:

$$\sum_{r=n+1}^{2n} r^2 = \sum_{r=1}^{n} (r+n)^2$$

$$= \sum_{r=1}^{n} (r^2 + 2nr + n^2)$$

$$= \frac{1}{6}n(n+1)(2n+1) + 2n \times \frac{1}{2}n(n+1) + n^3$$

$$= \frac{1}{6}n[(n+1)(2n+1) + 6n(n+1) + 6n^2]$$

$$= \frac{1}{6}n[2n^2 + 3n + 1 + 6n^2 + 6n + 6n^2]$$

$$= \frac{1}{6}n(14n^2 + 9n + 1)$$

$$= \frac{1}{6}n(7n+1)(2n+1)$$

30) c)

$$\sum_{r=1}^{n+4} r^2 = \frac{1}{6}(n+4)(n+4+1)(2(n+4)+1) = \frac{1}{6}(n+4)(n+5)(2n+9)$$

30) d)

$$\sum_{r=1}^{n+3}(r^3-2r^2)=\sum_{r=1}^{n+3}r^3-2\sum_{r=1}^{n+3}r^2$$

$$=\frac{1}{4}(n+3)^2(n+3+1)^2-2\times\frac{1}{6}(n+3)(n+3+1)(2(n+3)+1)$$

$$=\frac{1}{4}(n+3)^2(n+4)^2-\frac{1}{3}(n+3)(n+4)(2n+7)$$

$$=\frac{1}{12}(n+3)(n+4)[3(n+3)(n+4)-4(2n+7)]$$

$$=\frac{1}{12}(n+3)(n+4)[3n^2+21n+36-8n-28]$$

$$=\frac{1}{12}(n+3)(n+4)(3n^2+13n+8)$$

30) e)

$$\sum_{r=n+1}^{2n}(r^3+r)=\sum_{r=1}^{2n}(r^3+r)-\sum_{r=1}^{n}(r^3+r)$$

$$=\sum_{r=1}^{2n}r^3+\sum_{r=1}^{2n}r-\sum_{r=1}^{n}r^3-\sum_{r=1}^{n}r$$

$$=\frac{1}{4}(2n)^2(2n+1)^2+\frac{1}{2}(2n)\big((2n)+1\big)-\frac{1}{4}n^2(n+1)^2-\frac{1}{2}n(n+1)$$

$$=n^2(2n+1)^2+n(2n+1)-\frac{1}{4}n^2(n+1)^2-\frac{1}{2}n(n+1)$$

$$=\frac{1}{4}n[4n(2n+1)^2+4(2n+1)-n(n+1)^2-2(n+1)]$$

$$=\frac{1}{4}n[4n(4n^2+4n+1)+4(2n+1)-n(n^2+2n+1)-2(n+1)]$$

$$=\frac{1}{4}n[16n^3+16n^2+4n+8n+4-n^3-2n^2-n-2n-2]$$

$$=\frac{1}{4}n(15n^3+14n^2+9n+2)$$

$$=\frac{1}{4}n(3n+1)(5n^2+3n+2)$$

Alternative Method:

$$\sum_{r=n+1}^{2n}(r^3+r)=\sum_{r=1}^{n}\big((r+n)^3+(r+n)\big)=\sum_{r=1}^{n}(r^3+3nr^2+3n^2r+r+n+n^3)$$

$$=\frac{1}{4}n^2(n+1)^2+3n\times\frac{1}{6}n(n+1)(2n+1)+3n^2\times\frac{1}{2}n(n+1)+\frac{1}{2}n(n+1)+n^2+n^4$$

$$=n\left[\frac{1}{4}n(n+1)^2+\frac{1}{2}n(n+1)(2n+1)+\frac{3}{2}n^2(n+1)+\frac{1}{2}(n+1)+n+n^3\right]$$

$$=n\left[\frac{1}{4}n(n^2+2n+1)+\frac{1}{2}n(2n^2+3n+1)+\frac{3}{2}(n^3+n^2)+\frac{1}{2}(n+1)+n+n^3\right]$$

$$=n\left[\frac{1}{4}n^3+\frac{1}{2}n^2+\frac{1}{4}n+n^3+\frac{3}{2}n^2+\frac{1}{2}n+\frac{3}{2}n^3+\frac{3}{2}n^2+\frac{1}{2}n+\frac{1}{2}+n+n^3\right]$$

$$=n\left[\frac{15}{4}n^3+\frac{7}{2}n^2+\frac{9}{4}n+\frac{1}{2}\right]$$

$$=\frac{1}{4}n(15n^3+14n^2+9n+2)$$

$$=\frac{1}{4}n(3n+1)(5n^2+3n+2)$$

31)

$$\sum_{r=6}^{n+1} r^2 = \sum_{r=1}^{n+1} r^2 - \sum_{r=1}^{5} r^2$$
$$= \frac{1}{6}(n+1)(n+1+1)(2(n+1)+1) - \frac{1}{6} \times 5(5+1)(10+1)$$
$$= \frac{1}{6}(n+1)(n+2)(2n+3) - 55$$
$$= \frac{1}{6}[(n+1)(2n^2+7n+6) - 330]$$
$$= \frac{1}{6}[2n^3+9n^2+13n+6-330]$$
$$= \frac{1}{6}(2n^3+9n^2+13n-324)$$

32) a)

$$\sum_{r=1}^{n} r(r-5) = \sum_{r=1}^{n} r^2 - 5r = \sum_{r=1}^{n} r^2 - 5\sum_{r=1}^{n} r$$
$$= \frac{1}{6}n(n+1)(2n+1) - 5 \times \frac{1}{2}n(n+1)$$
$$= \frac{1}{6}n[(n+1)(2n+1) - 15(n+1)]$$
$$= \frac{1}{6}n[2n^2+3n+1-15n-15]$$
$$= \frac{1}{6}n(2n^2-12n-14)$$
$$= \frac{1}{3}n(n^2-6n-7)$$
$$= \frac{1}{3}n(n-7)(n+1)$$

32) b)

$$\sum_{r=n+1}^{3n} r(r-5) = \sum_{r=1}^{3n} r(r-5) - \sum_{r=1}^{n} r(r-5)$$
$$= \frac{1}{3}(3n)(3n-7)(3n+1) - \frac{1}{3}n(n-7)(n+1)$$
$$= \frac{1}{3}n[3(3n-7)(3n+1) - (n-7)(n+1)]$$
$$= \frac{1}{3}n[3(9n^2-18n-7) - (n^2-6n-7)]$$
$$= \frac{1}{3}n(27n^2-54n-21-n^2+6n+7)$$
$$= \frac{2}{3}n(13n^2-24n-7)$$

33) a)

$$\sum_{r=1}^{n}(2r)^2 = 4\sum_{r=1}^{n} r = 4 \times \frac{1}{6}n(n+1)(2n+1) = \frac{2}{3}n(n+1)(2n+1)$$

33) b)

$$\sum_{r=k}^{k+4}(2r)^2 = 1020$$

$$\sum_{r=k}^{k+4}(2r)^2 = \sum_{r=1}^{k+4}(2r)^2 - \sum_{r=1}^{k-1}(2r)^2$$

$$= \frac{2}{3}(k+4)(k+4+1)(2(k+4)+1) - \frac{2}{3}(k-1)(k-1+1)(2(k-1)+1)$$

$$\frac{2}{3}(k+4)(k+5)(2k+9) - \frac{2}{3}k(k-1)(2k-1) = 1020$$

$$(k+4)(k+5)(2k+9) - k(k-1)(2k-1) = 1530$$

$$(k+4)(2k^2+19k+45) - k(2k^2-3k+1) = 1530$$

$$2k^3+19k^2+45k+8k^2+76k+180-2k^3+3k^2-k = 1530$$

$$30k^2+120k-1350 = 0$$

which has solutions $k = -9$ or $k = 5$

Since k is positive, $k = 5$ so $2k = 10$

34)

$$\sum_{r=1}^{2n} r^3 = \frac{1}{4}(2n)^2(2n+1)^2 = n^2(4n^2+4n+1) = n(4n^3+4n^2+n)$$

$$\sum_{r=1}^{n}(r+2)^2 = \sum_{r=1}^{n}(r^2+4r+4) = \sum_{r=1}^{n} r^2 + 4\sum_{r=1}^{n} r + 4\sum_{r=1}^{n} 1$$

$$= \frac{1}{6}n(n+1)(2n+1) + 4 \times \frac{1}{2}n(n+1) + 4n$$

$$= \frac{1}{6}n[(n+1)(2n+1) + 12(n+1) + 24]$$

$$= \frac{1}{6}n[2n^2+3n+1+12n+12+24]$$

$$= \frac{1}{6}n(2n^2+15n+37)$$

$$\sum_{r=1}^{2n} r^3 - \sum_{r=1}^{n}(r+2)^2 = n(4n^3+4n^2+n) - \frac{1}{6}n(2n^2+15n+37)$$

$$= \frac{1}{6}n[24n^3+24n^2+6n-2n^2-15n-37]$$

$$= \frac{1}{6}n[24n^3+22n^2-9n-37]$$

$$= \frac{1}{6}n(n-1)(24n^2+46n+37)$$

35)

$$\sum_{r=1}^{n}(r+a)(r+b)=\sum_{r=1}^{n}(r^2+(a+b)r+ab)$$

$$=\sum_{r=1}^{n}r^2+(a+b)\sum_{r=1}^{n}r+ab\sum_{r=1}^{n}1$$

$$=\frac{1}{6}n(n+1)(2n+1)+(a+b)\times\frac{1}{2}n(n+1)+abn$$

$$=\frac{1}{3}n\left[\frac{1}{2}(n+1)(2n+1)+\frac{3}{2}(a+b)(n+1)+3ab\right]$$

$$=\frac{1}{3}n\left[n^2+\frac{3}{2}n+\frac{1}{2}+\frac{3}{2}(a+b)n+\frac{3}{2}(a+b)+3ab\right]$$

$$=\frac{1}{3}n\left[n^2+\left(\frac{3}{2}+\frac{3}{2}(a+b)\right)n+\frac{1}{2}+\frac{3}{2}(a+b)+3ab\right]$$

So comparing terms with $\frac{1}{3}n(n^2-7)$

$\frac{3}{2}+\frac{3}{2}(a+b)=0$

$\frac{1}{2}+\frac{3}{2}(a+b)+3ab=-7$

Rearrange the first:

$\frac{3}{2}(a+b)=-\frac{3}{2}\Rightarrow a=-b-1$

Sub into second:

$\frac{1}{2}+\frac{3}{2}(-b-1+b)+3(-b-1)b=-7$

$-1-3b^2-3b=-7$

$3b^2+3b-6=0$ which has solutions $b=-2$ or $b=1$

$\therefore b=-2, a=1$ or $a=-2, b=1$

36) a)

$$\sum_{r=1}^{2n}(n-r)^2=\sum_{r=1}^{2n}(n^2-2nr+r^2)=n^2\sum_{r=1}^{2n}1-2n\sum_{r=1}^{2n}r+\sum_{r=1}^{2n}r^2$$

$$=n^2(2n)-2n\times\frac{1}{2}(2n)(2n+1)+\frac{1}{6}(2n)(2n+1)(4n+1)$$

$$=2n^3-2n^2(2n+1)+\frac{1}{3}n(2n+1)(4n+1)$$

$$=\frac{1}{3}n[6n^2-6n(2n+1)+(2n+1)(4n+1)]$$

$$=\frac{1}{3}n[6n^2-12n^2-6n+8n^2+6n+1]$$

$=\frac{1}{3}n(2n^2+1)$ as req.

36) b)

$$\sum_{r=1}^{n}(n-r)^2 = n^2\sum_{r=1}^{n}1 - 2n\sum_{r=1}^{n}r + \sum_{r=1}^{n}r^2$$

$$= n^2 \times n - 2n \times \frac{1}{2}n(n+1) + \frac{1}{6}n(n+1)(2n+1)$$

$$= \frac{1}{6}n[6n^2 - 6n(n+1) + (n+1)(2n+1)]$$

$$= \frac{1}{6}n[6n^2 - 6n^2 - 6n + 2n^2 + 3n + 1]$$

$$= \frac{1}{6}n(2n^2 - 3n + 1)$$

So that

$$\sum_{r=n+1}^{2n}(n-r)^2 = \sum_{r=1}^{2n}(n-r)^2 - \sum_{r=1}^{n}(n-r)^2$$

$$= \frac{1}{3}n(2n^2+1) - \frac{1}{6}n(2n^2 - 3n + 1)$$

$$= \frac{1}{6}n[2(2n^2+1) - 2n^2 + 3n - 1]$$

$$= \frac{1}{6}n[4n^2 + 2 - 2n^2 + 3n - 1]$$

$$= \frac{1}{6}n[2n^2 + 3n + 1]$$

$$= \frac{1}{6}n(n+1)(2n+1) = \sum_{r=1}^{n}r^2$$

as req.

37) a)

$$\sum_{r=1}^{3n-2}(5r+3) = 5\sum_{r=1}^{3n-2}r + \sum_{r=1}^{3n-2}3$$

$$= 5\left(\frac{1}{2}(3n-2)(3n-2+1)\right) + 3(3n-2)$$

$$= \frac{5}{2}(9n^2 - 9n + 2) + 9n - 6$$

$$= \frac{1}{2}(45n^2 - 45n + 10) + \frac{1}{2}(18n - 12)$$

$$= \frac{1}{2}(45n^2 - 27n - 2)$$

$$= \frac{1}{2}(3n-2)(15n+1)$$

37) b)

$3n - 2 = 16 \Rightarrow n = 6$

$\frac{1}{2}(45 \times 6^2 - 27 \times 6 - 2) = 728$

38) a)

$$\sum_{r=1}^{2n+7}(r^2 - 1) = \sum_{r=1}^{2n+7} r^2 - \sum_{r=1}^{2n+7} 1$$

$= \frac{1}{6}(2n + 7)(2n + 7 + 1)(2(2n + 7) + 1) - (2n + 7)$

$= \frac{1}{6}(2n + 7)(2n + 8)(4n + 15) - (2n + 7)$

$= \frac{1}{3}(2n + 7)[(n + 4)(4n + 15) - 3]$

$= \frac{1}{3}(2n + 7)[4n^2 + 31n + 57]$

$= \frac{1}{3}(2n + 7)(4n + 19)(n + 3)$ as req.

38) b)

$2n + 7 = 63 \Rightarrow n = 28$

$\frac{1}{3}(2 \times 28 + 7)(4 \times 28 + 19)(28 + 3) = 85\,281$

$2n + 7 = 9 \Rightarrow n = 1$

$\frac{1}{3}(2 \times 1 + 7)(4 \times 1 + 19)(1 + 3) = 276$

$85\,281 - 276 = 85\,005$

39) a)

$$\sum_{r=1}^{8}(-1)^r = -1 + 1 - 1 + 1 - 1 + 1 - 1 + 1 = 4(-1) + 4(1) = 0$$

39) b)

$$\sum_{r=1}^{2n}(-1)^r = -1 + 1 - \cdots - 1 + 1 = n(-1) + n(1) = 0$$

39) c)

$$\sum_{r=1}^{9}(-1)^r = -1+1-1+1-1+1-1+1-1 = 5(-1)+4(1) = -1$$

39) d)

$$\sum_{r=1}^{2n-1}(-1)^r = -1+1-\cdots-1+1-1 = n(-1)+(n-1)(1) = -1$$

40) a)

$$\sum_{r=1}^{2n}(-1)^r r = -\sum_{r=1}^{n}(2r-1)+\sum_{r=1}^{n}(2r)$$

$$= -2\sum_{r=1}^{n} r + \sum_{r=1}^{n} 1 + 2\sum_{r=1}^{n} r$$

$$= \sum_{r=1}^{n} 1 = n$$

40) b)

$$\sum_{r=1}^{2n-1}(-1)^r r = -\sum_{r=1}^{n}(2r-1)+\sum_{r=1}^{n-1}(2r)$$

$$= -2\sum_{r=1}^{n} r + \sum_{r=1}^{n} 1 + 2\sum_{r=1}^{n-1} r$$

$$= -2\times\frac{1}{2}n(n+1)+n+2\times\frac{1}{2}(n-1)n$$

$$= -n^2-n+n+n^2-n$$

$$= -n$$

40) c)

$$\sum_{r=1}^{2n}(-1)^r(r+1) = -\sum_{r=1}^{n}(2r) + \sum_{r=1}^{n}(2r+1)$$

$$= -2\sum_{r=1}^{n} r + 2\sum_{r=1}^{n} r + \sum_{r=1}^{n} 1$$

$$= n$$

40) d)

$$\sum_{r=1}^{2n}(-1)^r r^2 = -\sum_{r=1}^{n}(2r-1)^2 + \sum_{r=1}^{n}(2r)^2$$

$$= -4\sum_{r=1}^{n} r^2 + 4\sum_{r=1}^{n} r - \sum_{r=1}^{n} 1 + 4\sum_{r=1}^{n} r^2$$

$$= 4\sum_{r=1}^{n} r - \sum_{r=1}^{n} 1$$

$$= 4 \times \frac{1}{2}n(n+1) - n$$

$$= 2n^2 + 2n - n$$

$$= n(2n+1)$$

41) a)

$$\sum_{r=1}^{10} \sin\left(\frac{\pi}{2}r\right) = \sin\frac{\pi}{2} + \sin\pi + \sin\frac{3\pi}{2} + \sin 2\pi + \cdots + \sin 5\pi$$

$$= 1 + 0 - 1 + 0 + 1 + 0 - 1 + 0 + 1 + 0$$

$$= 1$$

41) b)

$$\sum_{r=1}^{32} r\cos\left(\frac{\pi}{2}r\right) = \cos\frac{\pi}{2} + 2\cos\pi + 3\cos\frac{3\pi}{2} + 4\cos 2\pi + \cdots + 32\cos 5\pi$$

$$= 0 - 2 + 0 + 4 + 0 - 6 + 0 + 1 + \cdots + 0 - 30 + 0 + 32$$

$$= -2 + 4 - 6 + 8 + \cdots - 30 + 32$$

$$= 8 \times 2 = 16$$

41) c)

$$\sum_{r=1}^{27} r\tan\left(\frac{\pi}{3}r\right) = \tan\frac{\pi}{3} + 2\tan\frac{2\pi}{3} + 3\tan\pi + 4\tan\frac{4\pi}{3} + \cdots + 27\tan\pi$$
$$= \sqrt{3} - 2\sqrt{3} + 0 + 4\sqrt{3} - 6\sqrt{3} + \cdots + 25\sqrt{3} - 26\sqrt{3} + 0$$
$$= 9 \times -\sqrt{3} = -9\sqrt{3}$$

42) a)

$$\sum_{r=1}^{n} \sin(\pi r) = \sin\pi + \sin 2\pi + \sin 3\pi + \cdots + \sin n\pi$$
$$= 0$$

42) b)

$$\sum_{r=1}^{n} \cos(2\pi r) = \cos(2\pi) + \cos(4\pi) + \cos(6\pi) + \cdots + \cos(2\pi n)$$
$$= 1 + 1 + 1 + \cdots + 1$$
$$= n$$

42) c)

$$\sum_{r=1}^{3n} \tan\left(\frac{\pi}{3}r\right) = \tan\left(\frac{\pi}{3}\right) + \tan\left(\frac{2\pi}{3}\right) + \tan(\pi) + \cdots + \tan(n\pi)$$
$$= \sqrt{3} - \sqrt{3} + 0 + \cdots + \sqrt{3} - \sqrt{3} + 0$$
$$= 0$$

43) a)

$$\sum_{r=1}^{n}(2r+1)^2 = \sum_{r=1}^{n}(4r^2+4r+1)$$

$$= 4\sum_{r=1}^{n} r^2 + 4\sum_{r=1}^{n} r + \sum_{r=1}^{n} 1$$

$$= 4 \times \frac{1}{6}n(n+1)(2n+1) + 4 \times \frac{1}{2}n(n+1) + n$$

$$= \frac{1}{3}n[2(n+1)(2n+1) + 6(n+1) + 3]$$

$$= \frac{1}{3}n[4n^2 + 6n + 2 + 6n + 6 + 3]$$

$$= \frac{1}{3}n(4n^2 + 12n + 11) \text{ as req.}$$

43) b)

$$\sum_{r=1}^{k} \cos(2\pi r) = 1 + 1 + 1 + \cdots + 1 = k$$

$$\therefore 15\sum_{r=1}^{k} \cos(2\pi r) = 15k$$

$$\therefore \sum_{r=1}^{k}(2r+1)^2 + 15\sum_{r=1}^{k} \cos(2\pi r) = \frac{1}{3}k(4k^2+12k+11) + 15k$$

$$\frac{1}{3}k(4k^2+12k+11) + 15k = 24k^2 \Rightarrow \frac{4}{3}k^3 - 20k^2 + \frac{56}{3}k = 0$$

Which gives $k = 0$, $k = 1$ or $k = 14$

So $k = 1$ or $k = 14$

44) a)

$$\sum_{r=1}^{n}(5r-3)^2 = \sum_{r=1}^{n}(25r^2-30r+9)$$

$$= 25\sum_{r=1}^{n}r^2 - 30\sum_{r=1}^{n}r + 9\sum_{r=1}^{n}1$$

$$= 25\times\frac{1}{6}n(n+1)(2n+1) - 30\times\frac{1}{2}n(n+1) + 9n$$

$$= \frac{1}{6}n[25(n+1)(2n+1) - 90(n+1) + 54]$$

$$= \frac{1}{6}n[50n^2 + 75n + 25 - 90n - 90 + 54]$$

$$= \frac{1}{6}n(50n^2 - 15n - 11) \text{ as req.}$$

44) b)

$$\sum_{r=1}^{20} r\sin\left(\frac{\pi}{2}r\right)$$

$$= 1\times 1 + 2\times 0 + 3\times -1 + 4\times 0 + \cdots + 17\times 1 + 18\times 0 + 19\times -1 + 20\times 0$$

$$= 1 - 3 + \cdots + 17 - 19 = 5\times -2 = -10$$

$$\therefore 20\sum_{r=1}^{20} r\sin\left(\frac{\pi}{2}r\right) = 20\times -10$$

$$\sum_{r=3}^{k}(5r-3)^2 = \frac{1}{6}k(50k^2-15k-11) - 4 - 49 = \frac{25}{3}k^3 - \frac{5}{2}k^2 - \frac{11}{6}k - 53$$

$$\therefore \sum_{r=3}^{k}(5r-3)^2 + 20\sum_{r=1}^{20} r\sin\left(\frac{\pi}{2}r\right) = \frac{25}{3}k^3 - \frac{5}{2}k^2 - \frac{11}{6}k - 253$$

$$\frac{25}{3}k^3 - \frac{5}{2}k^2 - \frac{11}{6}k - 253 = 241k \Rightarrow \frac{25}{3}k^3 - \frac{5}{2}k^2 - \frac{1457}{6}k - 253 = 0$$

Which gives $k = -\frac{23}{5}$, $k = -\frac{11}{10}$ or $k = 6$ So $k = 6$

45) a)

$$\sum_{r=1}^{2n} r((-1)^r + 5)^2 = \sum_{r=1}^{2n} r(1 + (-1)^r 10 + 25) = \sum_{r=1}^{2n} (26r + (-1)^r 10r)$$

$$= 26\sum_{r=1}^{2n} r - 10\sum_{r=1}^{n} (2r - 1) + 10\sum_{r=1}^{n} (2r)$$

$$= 26\sum_{r=1}^{2n} r - 20\sum_{r=1}^{n} r + 10\sum_{r=1}^{n} 1 + 20\sum_{r=1}^{n} r$$

$$= 26\sum_{r=1}^{2n} r + 10\sum_{r=1}^{n} 1$$

$$= 26 \times \frac{1}{2}(2n)(2n + 1) + 10n$$

$$= 26n(2n + 1) + 10n$$

$$= 52n^2 + 26n + 10n$$

$$= 52n^2 + 36n$$

$$= n(52n + 36)$$

45) b)

$$40(52 \times 40 + 36) - 7(52 \times 7 + 36) = 81\,840$$

46) a)

$$\sum_{r=1}^{2n}((-1)^r+3r)^2=\sum_{r=1}^{2n}(1+(-1)^r6r+9r^2)$$

$$=\sum_{r=1}^{2n}1-6\sum_{r=1}^{n}(2r-1)+6\sum_{r=1}^{n}(2r)+9\sum_{r=1}^{2n}r^2$$

$$=\sum_{r=1}^{2n}1-12\sum_{r=1}^{n}r+6\sum_{r=1}^{n}1+12\sum_{r=1}^{n}r+9\sum_{r=1}^{2n}r^2$$

$$=\sum_{r=1}^{2n}1+6\sum_{r=1}^{n}1+9\sum_{r=1}^{2n}r^2$$

$$=2n+6n+9\times\frac{1}{6}\times2n(2n+1)(2(2n)+1)$$

$$=2n+6n+3n(2n+1)(4n+1)$$

$$=2n+6n+3n(8n^2+6n+1)$$

$$=2n+6n+24n^3+18n^2+3n$$

$$=24n^3+18n^2+11n$$

46) b)

$$24\times50^3+18\times50^2+11\times50-(24\times24^3+18\times24^2+11\times24)$$

$$=2\,703\,142$$